

When Retail Demand Moves Transit Supply: Spatial Price Competition with a Fixed Fleet

Abstract

We study how demand-responsive transport supply changes spatial retail price competition. Two retail destinations compete in prices, while a third-party operator reallocates a fixed service resource across directions in response to passenger demand. One direction also carries exogenous background travel demand. A retailer's price therefore affects its market share directly and indirectly by shifting service toward its own direction and away from its rival. We show that a nonempty open set of global pure-strategy Hotelling–Bertrand equilibria exhibits one-sided strategic substitutability: one retailer lowers its price when the rival raises price, while the other retains an upward-sloping response. A minimum service obligation can be strictly slack at equilibrium yet support the global equilibrium by truncating extreme off-path service withdrawal. The operator's waiting-cost minimization also yields an exact welfare mapping between the access term in shopper demand and marginal aggregate waiting cost. The decentralized allocation generally differs from the constrained-efficient allocation under the same service floor. The mechanism is locally robust to power waiting-cost specifications and applies most naturally to managed directional, shuttle, and demand-responsive fleet-allocation settings.

Keywords: spatial competition; transport economics; retail pricing; service frequency; fixed fleet; strategic interaction

JEL classification: L13; L91; R41

1 Introduction

How does spatial price competition change when a destination's demand can move transport service toward that destination and away from its rival? Standard spatial-competition models usually treat access as fixed. In many managed transport systems, however, service intensity responds to passenger demand while the total service resource is scarce. A destination that attracts more trips can then receive more service only by drawing that resource away from another direction or zone. Retail pricing affects not only who shops where, but also the access conditions under which the two destinations compete.

We study this feedback in a Hotelling–Bertrand model with two endpoint retail destinations and a third-party transport operator. The operator reallocates a fixed total service resource across the two directions in response to total passenger demand. One direction also carries exogenous background demand, such as commuting or education trips. A retailer's price cut raises its shopping demand, shifts service toward its own direction, withdraws service from the rival direction, and thereby feeds back into shopping demand. Directional background demand makes this shared-resource feedback asymmetric.

The main result is that the feedback can reverse the strategic relation for only one retailer. There exists a nonempty open set of primitives for which the model has a global pure-strategy price

Nash equilibrium satisfying

$$\frac{dBR_L}{dp_R} < 0 < \frac{dBR_R}{dp_L}.$$

Thus one retailer lowers its price when the rival raises price, while the other retains the conventional upward-sloping response. Strategic asymmetry itself is known (Tombak, 2006), and mixed strategic relations also arise in transport-network pricing (McHardy, 2024). Our contribution is narrower: the sign pattern is generated by downstream demand moving a third party’s shared transport resource, and it survives the global retail-price game rather than appearing only in local first-order conditions.

The minimum service obligation has a separate global role. Without a lower service bound, a sufficiently large price-induced demand shift can drive service in the losing direction toward zero and create multiple fulfilled-expectations continuations, a form of demand–frequency feedback familiar in transit economics (Bar-Yosef et al., 2013). We show that a minimum service share can be strictly slack at the equilibrium and yet support that equilibrium by truncating only sufficiently large off-path reallocations. For the exact existence witness, the same on-path equilibrium is supported over an intermediate range of nonbinding floor levels. We do not interpret that range as an optimal regulatory rule.

The transport microfoundation also disciplines welfare accounting. Conditional on retail shares, the operator minimizes aggregate waiting cost. The resulting envelope identity makes the directional waiting-cost difference faced by the marginal shopper equal to the derivative of minimized aggregate waiting cost. Retail prices cancel as transfers in welfare, so the decentralized allocation can be compared directly with the constrained-efficient allocation under the same service floor. At the exact witness, the decentralized equilibrium assigns 57.5% of shoppers to the high-background-demand side, versus approximately 70.24% in the same-floor constrained-efficient allocation, reducing real spatial and waiting costs by about 1.22% under the normalization. This is a welfare implication of the transport mechanism, not a claim that envelope arguments or private–social wedges are new in general.

Three nested benchmarks identify the interaction. The opposite-signed local price reactions disappear if service is fixed, if directional background demand is removed, or if service does not respond to shopping demand. The result is also locally robust when waiting disutility is generalized from w/f to $wf^{-\rho}$ around $\rho = 1$. The exact parameter vector is a constructive existence device; the economic claim is the open-set mechanism, not the particular witness.

The model is intended for settings in which downstream destinations are large enough to affect discretionary trip demand and service allocation. Directional shuttle or fixed-route operations using deadheading or short-turning provide one interpretation (Furth, 1985; Cortés et al., 2011); demand-responsive or shared-mobility fleets repositioned across zones provide another (Liu and Ouyang, 2021). The operator is a service-planning layer with a fixed effective resource, not a private platform whose fare, fleet size, or network design is modeled. We therefore do not claim that an ordinary small retailer changes public-transit frequency or that the model describes a generic balanced two-way timetable.

The paper sits at the intersection of spatial competition, network externalities, vertical transport markets, capacity allocation, and transit service design. Market-share feedback in Hotelling competition is well established (Grilo et al., 2001; Griva and Vettas, 2011; Tolotti and Yezpe, 2020; Toshimitsu, 2026); upstream transport facilities and service quality have long been studied jointly with downstream competition (Basso and Zhang, 2007; de Palma et al., 2018; Álvarez-Sanj Jaime et al., 2020); and scarce upstream capacity allocation is known to reshape downstream rivalry (Chen et al., 2013). The remaining claim is the complete transport-specific chain in which downstream demand reallocates one common service resource across rival directions, altering both own and rival access and generating the global price equilibrium above. Section 7 develops this positioning.

The remainder of the paper is organized as follows. Section 2 presents the model and operator allocation. Section 3 derives the price-game mechanism and nested benchmarks. Section 4 establishes the global equilibrium and the off-path role of the service obligation. Section 5 develops the welfare mapping. Section 6 discusses robustness, institutional interpretation, and empirical implications. Section 7 positions the paper against the closest literatures, and Section 8 concludes.

2 Model

We consider two retail destinations located at the endpoints of a unit interval. Destination L is located at 0 and destination R at 1. A unit mass of shoppers is uniformly distributed on $z \in [0, 1]$. The endpoint normalization follows the quadratic spatial-competition benchmark of d'Aspremont et al. (1979). Retailers have a common constant marginal cost c and choose prices p_L and p_R simultaneously.

The distinguishing feature is that access quality is supplied by a third-party transport operator. The operator has a fixed total service resource $F > 0$ that can be reallocated across the two directions. In addition to shoppers, there is an exogenous mass $M \geq 0$ of background passengers traveling toward the L direction. The background demand can represent commuting, education, or other travel that is independent of the retail price game.

2.1 Shopper choice

Let $t > 0$ be the coefficient on quadratic spatial mismatch and let $w > 0$ scale waiting disutility. Shoppers have inelastic unit demand over the two destinations: each shopper chooses either L or R , and no outside option is modeled. The common gross value v is therefore a welfare normalization and drops out of destination choice. If the operator supplies directional service levels f_L and f_R , the utilities of a shopper located at z are

$$U_L(z) = v - p_L - tz^2 - \frac{w}{f_L}, \quad (1)$$

$$U_R(z) = v - p_R - t(1 - z)^2 - \frac{w}{f_R}. \quad (2)$$

Let $x \in [0, 1]$ denote the shopper share served by L . The marginal shopper satisfies

$$p_L - p_R + t(2x - 1) + \frac{w}{f_L} - \frac{w}{f_R} = 0. \quad (3)$$

Retail profits are

$$\pi_L = (p_L - c)x, \quad \pi_R = (p_R - c)(1 - x). \quad (4)$$

2.2 Third-party service allocation

Conditional on the shopper share x , total directional passenger demands are

$$D_L = M + x, \quad D_R = 1 - x. \quad (5)$$

The operator allocates the fixed resource F to minimize aggregate waiting disutility subject to a minimum service requirement:

$$\min_{f_L, f_R} \quad w \left(\frac{M + x}{f_L} + \frac{1 - x}{f_R} \right), \quad (6)$$

$$\text{s.t.} \quad f_L + f_R = F, \quad f_L, f_R \geq \underline{f}. \quad (7)$$

Define the minimum service share

$$q \equiv \frac{f}{F} \in (0, 1/2). \quad (8)$$

Absent the floor, the unique cost-minimizing share of service allocated to direction L is

$$s^u(x, M) = \frac{\sqrt{M+x}}{\sqrt{M+x} + \sqrt{1-x}}. \quad (9)$$

With the service obligation, the implemented service share is the clipped rule

$$s_q(x, M) = \min\{1 - q, \max\{q, s^u(x, M)\}\}, \quad (10)$$

so that

$$f_L = F s_q(x, M), \quad f_R = F[1 - s_q(x, M)]. \quad (11)$$

Let

$$A \equiv \frac{w}{F}, \quad h_q(x, M) \equiv \frac{1}{s_q(x, M)} - \frac{1}{1 - s_q(x, M)}. \quad (12)$$

Substituting the operator response into (3), the fulfilled-expectations shopper condition becomes

$$G_q(x; p_L, p_R) \equiv p_L - p_R + t(2x - 1) + A h_q(x, M) = 0. \quad (13)$$

When the service floor is slack,

$$h_q(x, M) = H(x, M) \equiv \frac{1 - M - 2x}{\sqrt{(M+x)(1-x)}}. \quad (14)$$

The fixed-resource constraint is central: an increase in service toward one direction necessarily withdraws service from the other.

2.3 Timing and equilibrium concept

Retailers first choose prices simultaneously. Given a price pair, the shopper share and operator allocation are then determined jointly as a fulfilled-expectations continuation. Formally, for any candidate share x the operator solves (6); shoppers choose between the two destinations taking the resulting service levels as given; and a continuation is a share that is consistent with those choices through (13), with the usual corner allocation if its zero lies outside $[0, 1]$. This is a fixed-point description of the continuation, not a claim that the operator literally observes realized shopping demand before shoppers choose.

Retailers anticipate this continuation mapping when setting prices. We study pure-strategy Nash equilibrium in retail prices with the shopper/operator continuation solved globally. A pair of local price first-order conditions is therefore not sufficient for equilibrium.

The canonical parameter domain is

$$t > 0, \quad F > 0, \quad w > 0, \quad M \geq 0, \quad q \in (0, 1/2), \quad (15)$$

with inelastic unit shopper demand. The main result is an existence result on a nonempty open subset of this domain, not a claim that the same equilibrium structure obtains for all admissible primitives.

3 Price competition and the service-allocation feedback

This section isolates the local pricing mechanism before imposing the global-deviation requirement in Section 4. Throughout the interior slack-service region, define

$$g(x, M) \equiv G_{q,x} = 2t + AH_x(x, M), \quad g'(x, M) = AH_{xx}(x, M). \quad (16)$$

Direct differentiation of (14) gives

$$H_x(x, M) = -\frac{(M+1)^2}{2[(M+x)(1-x)]^{3/2}} < 0, \quad (17)$$

$$H_{xx}(x, M) = -\frac{3(M+1)^2(M+2x-1)}{4[(M+x)(1-x)]^{5/2}}. \quad (18)$$

A stable fulfilled-expectations branch requires $g > 0$. On such a branch,

$$\frac{\partial x}{\partial p_L} = -\frac{1}{g}, \quad \frac{\partial x}{\partial p_R} = \frac{1}{g}. \quad (19)$$

Because $H_x < 0$, demand-responsive service allocation makes demand more price-sensitive than under fixed service whenever the floor is slack.

3.1 Retail first-order conditions

Using (19), the interior retail first-order conditions are

$$p_L - c = xg, \quad p_R - c = (1-x)g. \quad (20)$$

Combining (20) with shopper indifference yields the equilibrium-share equation

$$(2x-1)g + t(2x-1) + AH(x, M) = 0. \quad (21)$$

The own-price second-order conditions can be written as

$$S_L \equiv 2g + xg' > 0, \quad S_R \equiv 2g - (1-x)g' > 0. \quad (22)$$

Implicit differentiation of the two first-order conditions gives the local best-response slopes

$$\frac{dBR_L}{dp_R} = \frac{g + xg'}{2g + xg'}, \quad (23)$$

$$\frac{dBR_R}{dp_L} = \frac{g - (1-x)g'}{2g - (1-x)g'}. \quad (24)$$

Thus the service-allocation feedback affects strategic interaction through the curvature $g' = AH_{xx}$. Directional background demand shifts that curvature asymmetrically across the market.

Lemma 1 (Local one-sided strategic substitutability). *On a stable slack-service branch satisfying (22), L treats R 's price as a strategic substitute while R treats L 's price as a strategic complement whenever*

$$g + xg' < 0 \quad \text{and} \quad g - (1-x)g' > 0. \quad (25)$$

Equivalently, the first inequality requires sufficiently negative curvature of the demand-mediated service effect on the L side, while the second preserves conventional complementarity on the R side.

The terminology follows the general concept of strategic asymmetry in Tombak (2006); the contribution here is not the sign pattern itself but its third-party transport microfoundation and its survival in a global retail-price equilibrium.

3.2 Nested benchmarks

The interaction can be separated from three simpler cases.

Proposition 1 (Nested interaction identification). *The opposite-signed reaction pattern in Lemma 1 disappears in each of the following minimum benchmarks.*

1. *If directional service is fixed with respect to retail demand, then $g' = 0$ and both local price-reaction slopes equal $1/2$.*
2. *If $M = 0$, demand-responsive service preserves a symmetric equilibrium at $x = 1/2$. At that point $H_{xx} = 0$, so both reaction slopes again equal $1/2$. The symmetric markup is $p - c = t - 2w/F$ when the interior stable branch exists.*
3. *If $M > 0$ but service allocation does not respond to shopping demand, background access asymmetry can shift market shares and price levels, but $g' = 0$ in the retail demand response and both local reaction slopes remain $1/2$.*

Proposition 1 is an identification result, not a separate novelty claim. In particular, network or clientele effects in spatial price competition are well established (Grilo et al., 2001; Griva and Vettas, 2011; Tolotti and Yepez, 2020). The point is that neither directional background demand nor demand-responsive service alone produces the one-sided reaction pattern in the minimum nested comparisons.

4 Global equilibrium and a slack service obligation

The local first-order conditions in Section 3 are not sufficient for equilibrium. Without a lower service bound, a large retail-demand shift can drive service in the losing direction toward zero. The resulting demand–frequency feedback can create multiple fulfilled-expectations continuations and profitable branch-jump deviations. We do not claim equilibrium-selection-independent nonexistence when the floor is removed. Such demand–frequency vicious and virtuous cycles are known in transit systems (Bar-Yosef et al., 2013); we use the service floor only to obtain a globally well-defined continuation while preserving the local feedback at the candidate equilibrium.

4.1 An exact global-equilibrium witness

The existence proof uses the normalization

$$t = F = 1, \quad c = 0, \tag{26}$$

and the primitives

$$M = \frac{2}{3}, \quad q = \frac{1}{3}, \quad x^* = \frac{23}{40}, \quad \frac{w}{Ft} = \frac{22797\sqrt{7599}}{7364680}. \tag{27}$$

At this point the unconstrained service share is

$$s^* = \frac{\sqrt{149}}{\sqrt{149} + \sqrt{51}} \approx 0.630895, \tag{28}$$

so both minimum-service constraints are strictly slack. For $M = 2/3$ and $q = 1/3$, the R -side floor begins to bind at $x = 2/3$; over the binding region the waiting-cost difference is constant and $G_{q,x} = 2$.

Table 1: Exact global-equilibrium witness used in the existence proof

Object	Exact value	Decimal
M	$2/3$	0.666667
q	$1/3$	0.333333
x^*	$23/40$	0.575000
$w/(Ft)$	$22797\sqrt{7599}/7364680$	0.269837
p_L^*	$2164691/3682340$	0.587857
p_R^*	$1599989/3682340$	0.434503
π_L^*	$49787893/147293600$	0.338018
π_R^*	$27199813/147293600$	0.184664
BR_L'	$-65891639/172506722$	-0.381966
BR_R'	$27253433/41276866$	0.660259

On the slack region $x \leq 2/3$,

$$g(x) = 2 + AH_x(x, 2/3). \quad (29)$$

The minimum of this slope on the relevant interval is attained at the floor boundary and equals

$$g(2/3) = \frac{47133952 - 341955\sqrt{7599}}{23566976} \approx 0.735137 > 0. \quad (30)$$

Hence G_q is strictly increasing over the whole consumer interval. Every unilateral price choice therefore induces a unique shopper/operator continuation, with corner market shares when the unique zero lies outside $[0, 1]$.

At the witness, the own-price second-order conditions are strict and the two best-response slopes have opposite signs, as reported in Table 1. Global optimality is established by reparameterizing each retailer's unilateral price problem by the uniquely induced share x . For L , the only competing maximum outside x^* is the service-floor boundary, and the exact profit gap is

$$\pi_L(x^*) - \pi_L(2/3) = \frac{358680877 - 4103460\sqrt{7599}}{1325642400} \approx 0.000734 > 0. \quad (31)$$

For R , the best deviation in the binding-floor region is the vertex of a concave quadratic and remains below the candidate profit by approximately 0.009698. Exact root isolation also verifies that no interior stationary point dominates x^* .

Proposition 2 (Global network-mediated strategic asymmetry). *There exists a nonempty open set of primitives for which the model has a global pure-strategy retail-price Nash equilibrium satisfying*

$$\frac{dBR_L}{dp_R} < 0 < \frac{dBR_R}{dp_L}. \quad (32)$$

The proof is constructive. Equations (27)–(31) provide an exact witness with a globally single-valued continuation, strict own-price second-order conditions, strict opposite reaction signs, and strict global-deviation profit gaps for both firms. All decisive inequalities are strict, so continuity yields a nonempty open neighborhood with the same equilibrium structure. A deterministic audit of a nearby parameter grid finds 595 candidates satisfying the local conditions, of which 444 also pass the monotone-continuation and global-deviation checks; this audit supports but does not replace the exact argument.

4.2 Why a nonbinding service floor can matter globally

The service obligation is not the source of Proposition 2. At the equilibrium it is slack, so neither the local demand slope nor the local price first-order conditions depend on q . Its role is off path: sufficiently large price deviations can otherwise trigger extreme service withdrawal from the rival direction.

Proposition 3 (Slack service obligation and global support). *A minimum service obligation can be strictly slack at a price equilibrium satisfying Proposition 2 while supporting that equilibrium by truncating only sufficiently extreme off-equilibrium service reallocations.*

For the exact witness, the same on-path prices, market share, frequencies, and welfare are supported over the intermediate interval

$$q_L \approx 0.324091 < q < q_U \approx 0.344228, \quad (33)$$

while the floor remains slack at the candidate. For q just below q_L , while the continuation remains monotone, L can cut price enough to exploit the high- L continuation before the floor activates. If $q > q_U$ while still slack at the candidate, the off-path protection of R becomes strong enough to create a profitable R deviation. Thus a regulation that is nonbinding in normal operation can be too weak or too strong solely through its effect on off-equilibrium continuations.

Equation (33) is witness-specific. We do not interpret q_L and q_U as universal policy thresholds, nor do we solve for an optimal service standard. Minimum service itself is a standard transit-service primitive; the result here concerns its nonbinding-on-path role in the downstream global price game.

5 Welfare

The operator microfoundation provides a direct welfare interpretation of the same access feedback that shapes price competition. Let $\bar{v}$ denote the gross value of a trip for each background passenger. Shopper consumer surplus is

$$CS_S = \int_0^x \left[v - p_L - tz^2 - \frac{w}{f_L} \right] dz + \int_x^1 \left[v - p_R - t(1-z)^2 - \frac{w}{f_R} \right] dz \quad (34)$$

$$= v - p_L x - p_R(1-x) - \frac{t}{3} \{x^3 + (1-x)^3\} - A \left\{ \frac{x}{s_q} + \frac{1-x}{1-s_q} \right\}. \quad (35)$$

Background-passenger surplus is

$$CS_B = M\bar{v} - A \frac{M}{s_q}, \quad (36)$$

and retail producer surplus is

$$PS = (p_L - c)x + (p_R - c)(1-x). \quad (37)$$

Adding these terms cancels retail prices exactly. Up to constants, total welfare is therefore the negative of real spatial and waiting costs:

$$TW = v + M\bar{v} - c - C_q(x), \quad (38)$$

where

$$C_q(x) = \frac{t}{3} \{x^3 + (1-x)^3\} + A \left\{ \frac{M+x}{s_q(x, M)} + \frac{1-x}{1-s_q(x, M)} \right\}. \quad (39)$$

Any fixed cost of supplying the resource F is common across the comparisons considered here and can be added as a constant.

5.1 The operator envelope identity

When the service floor is slack, minimized aggregate waiting cost is

$$J(x) = A \left(\sqrt{M+x} + \sqrt{1-x} \right)^2. \quad (40)$$

The operator envelope theorem gives

$$J'(x) = AH(x, M). \quad (41)$$

Consequently,

$$C'(x) = t(2x - 1) + AH(x, M), \quad (42)$$

and the marginal shopper condition (13) can be written on the slack branch as

$$p_L - p_R + C'(x) = 0. \quad (43)$$

Proposition 4 (Operator-envelope welfare identity). *When the minimum service constraint is slack, the directional waiting-cost difference entering shopper choice equals the derivative of minimized aggregate waiting cost. Hence the access term that determines retail demand also has an exact real-cost counterpart in welfare.*

This identity is the main reason to retain the transport operator rather than replace it with an arbitrary market-share externality. A reduced-form externality can reproduce local demand curvature, but it does not by itself determine the associated resource or waiting-cost accounting. We use the identity to discipline the welfare interpretation of the pricing mechanism; the envelope theorem itself is not claimed as a general theoretical novelty.

5.2 Private and social allocation

On a stable slack branch, $C''(x) = g(x) > 0$. Combining (43) with the price first-order conditions (20) gives

$$C'(x^{NE}) = -(2x^{NE} - 1)C''(x^{NE}). \quad (44)$$

Proposition 5 (Private–social share wedge). *At a stable interior price equilibrium with a slack service floor, if $x^{NE} > 1/2$ then $C'(x^{NE}) < 0$, so a marginal reallocation of shoppers toward L lowers real social cost. If $x^{NE} < 1/2$, the symmetric statement holds for R. The local wedge vanishes at $x^{NE} = 1/2$.*

The decentralized price game therefore dampens the socially valuable concentration toward the direction that already carries greater total demand. The economic force is the service-frequency scale economy generated by sharing a fixed fleet across shoppers and background passengers. This is a welfare implication of the operator-mediated mechanism, not a claim that private–social allocation wedges are new in differentiated markets.

At the exact witness, $x^{NE} = 0.575$ and

$$C'(x^{NE}) = -\frac{282351}{1841170} \approx -0.153354. \quad (45)$$

Under the same service floor $q = 1/3$, the right-direction floor binds for sufficiently high x . In that region the unique real-cost minimizer is

$$x^{SB}(q) = \frac{1}{2} + \frac{A(1-2q)}{2tq(1-q)}. \quad (46)$$

Table 2: Witness-specific welfare and service-floor quantities

Object	Value
Decentralized share x^{NE}	0.575000
Same-floor second-best share x^{SB}	0.702378
Real cost at x^{NE}	0.930726
Real cost at x^{SB}	0.919348
Real-cost improvement	0.011378
Lower support threshold q_L	0.324091
Upper support threshold q_U	0.344228

For the witness,

$$x^{SB} \approx 0.702378. \quad (47)$$

The associated reduction in real cost is approximately 0.011378, about 1.22% of decentralized real cost under the witness normalization.

We call this the *same-floor constrained-efficient allocation* (or same-floor second best): the planner minimizes real cost over the shopper allocation while taking the exogenous service obligation q and the operator's allocation rule as given. It is an allocation benchmark, not a claim that the model supplies a policy instrument that directly assigns shoppers across destinations.

If the service obligation is removed from the planner problem at the witness, the utilitarian first best places all shoppers on the L side and shuts down service in the other direction. We do not interpret that corner as a policy recommendation. Rather, it shows that the service floor embodies a coverage or service-availability objective that is exogenous to the utilitarian objective in (38). Accordingly, the model does not derive an optimal coverage standard or attach an equity value to minimum service.

6 Robustness and institutional interpretation

6.1 Waiting-cost technology

The canonical model uses $a(f) = w/f$ because it delivers a transparent square-root allocation. The mechanism is not specific to that functional form. Let

$$a(f) = wf^{-\rho}, \quad \rho > 0. \quad (48)$$

For directional demands D_L and D_R , the unconstrained operator solution is

$$f_i = F \frac{D_i^{1/(\rho+1)}}{D_L^{1/(\rho+1)} + D_R^{1/(\rho+1)}}. \quad (49)$$

The minimized waiting cost is

$$J_\rho(D_L, D_R) = \frac{w}{F\rho} \left(D_L^{1/(\rho+1)} + D_R^{1/(\rho+1)} \right)^{\rho+1}, \quad (50)$$

and the envelope identity generalizes to

$$\frac{dJ_\rho}{dx} = a(f_L^*) - a(f_R^*). \quad (51)$$

All decisive equilibrium inequalities at $\rho = 1$ are strict, so the global equilibrium and opposite reaction signs are locally robust by continuity. A deterministic audit checks the continuation of the witness at sampled values in the conservative range $0.95 \leq \rho \leq 1.01$. We treat that numerical interval only as supporting evidence, not as an analytic threshold.

6.2 Transport interpretations

The operator in the model should be interpreted as a service planner or operational allocation layer that minimizes passenger waiting cost conditional on a fixed service resource. It is not a profit-maximizing transport platform choosing fares, fleet size, or network design. Likewise, F is an effective service resource: when literal vehicle circulation matters, the normalization $f_L + f_R = F$ is most natural when directional cycle-time requirements have already been absorbed into the service units. Endogenizing operator pricing, operating cost, fleet size, or unequal cycle times would change the game and is outside the present model.

The model is most naturally interpreted as competition between large destinations or commercial clusters whose discretionary demand is material for service planning. It should not be read literally as an ordinary small retailer changing the timetable of a conventional balanced two-way bus route.

Two transport settings provide direct institutional interpretations. First, directional fixed-route or shuttle operations can use deadheading or short-turning to concentrate effective passenger service in the higher-demand direction. Furth (1985) studies directional passenger imbalance and shows how deadheading can reallocate effective service with a fixed fleet, including waiting-time minimization. Cortés et al. (2011) integrate short-turning and deadheading in transit-service optimization and explicitly allow service concentration in high-demand portions or directions.

Second, demand-responsive transit and shared-mobility systems routinely reposition finite vehicle fleets across zones. Liu and Ouyang (2021) jointly optimize transit and demand-responsive services, including fleet size and repositioning operations. These settings motivate the physical possibility of reallocating a scarce vehicle resource across zones; they should not be read as evidence that a private platform necessarily solves the particular waiting-cost minimization problem in (6).

The minimum service restriction also has an institutional counterpart. U.S. Federal Transit Administration guidance requires fixed-route providers to adopt quantitative service standards including vehicle headway or service frequency, and notes that headway standards can establish minimum frequencies and respond to vehicle loads (Federal Transit Administration, 2012). We therefore treat q as an exogenous coverage or service-availability obligation. The paper does not derive the optimal standard.

6.3 Empirical implications

Although the paper is theoretical, the mechanism implies several qualitative empirical patterns. A destination-side demand shock should reallocate service only when supply is demand-responsive; with a fixed total fleet, service gained by one direction should be accompanied by service loss or longer waits in the rival direction. The feedback should weaken as F increases because $A = w/F$ falls. A binding service floor creates a kink: further demand shifts can no longer withdraw service from the protected direction. Finally, stronger directional background demand shifts market share and relative prices toward the high-background-demand side near the symmetric benchmark.

The sharpest prediction is strategic rather than purely allocative: sufficiently strong directional background demand can make the high-background-demand retailer or cluster lower its own price when the rival raises price, while the rival retains the conventional upward-sloping price response.

Testing that prediction would require a setting in which destination demand plausibly affects operator allocation; establishing such an empirical application is outside the present paper.

7 Related literature

The paper combines familiar ingredients, so the relevant comparison is not whether any one ingredient is new, but whether the complete strategic chain is already contained in an existing model.

The retail side is standard spatial price competition. We use the quadratic endpoint benchmark associated with Hotelling (1929) and d’Aspremont et al. (1979); retailer locations are fixed and no contribution is claimed for the transport-cost technology. The closest reduced-form threat instead comes from differentiated-price models with network or clientele effects. In that literature, market-share feedback can intensify competition, generate multiplicity, and produce asymmetric outcomes (Grilo et al., 2001; Griva and Vettas, 2011; Tolotti and Yepez, 2020; Toshimitsu, 2026). Our local retail demand system can likewise be written as a nonlinear share-dependent term. We therefore do not treat market-share feedback, multiplicity, or asymmetry as novel. Nor is the sign pattern itself new: Tombak (2006) formalizes strategic asymmetry, and transport-network pricing can contain complement and substitute forces simultaneously (McHardy, 2024).

A second comparison concerns vertical transport markets and scarce upstream resources. Transportation economics already studies how infrastructure charges, service quality, frequency, and capacity interact with downstream carrier pricing and competition (Basso and Zhang, 2007; de Palma et al., 2018; Álvarez-Sanjaime et al., 2020). More generally, upstream capacity-allocation rules can reshape downstream retailer competition (Chen et al., 2013). These precedents rule out any claim that an upstream transport layer or scarce resource affecting downstream prices is itself new. The present mechanism differs in the continuation: the upstream operator does not choose an access price or expand capacity, but reallocates one fixed shared service resource in response to downstream trip demand. A demand gain therefore changes generalized access costs on both sides of the downstream market.

A third comparison is transit service design. The demand–frequency feedback is a classic scale-economy mechanism (Mohring, 1972), and vicious or virtuous cycles with multiple equilibria have been analyzed directly (Bar-Yosef et al., 2013). Minimum service requirements, fixed-fleet constraints, and directional service reallocation are likewise familiar operations primitives. Deadheading and short-turning can concentrate effective service when directional demand is imbalanced (Furth, 1985; Cortés et al., 2011), while demand-responsive systems make finite-fleet allocation and repositioning across zones explicit (Liu and Ouyang, 2021). The paper uses these ideas as an institutional microfoundation rather than as separate novelty claims.

What remains is the interaction across the three layers: Retail prices determine shopping demand, which reallocates the third-party shared service resource. The resulting changes in own and rival access feed back into shopping demand and retail best responses.

Directional background demand makes this chain asymmetric, while a minimum service floor can remain slack on path and alter only large off-path reallocations. The prior-art audit did not identify a single predecessor that combines this demand-responsive third-party continuation with a global Hotelling–Bertrand equilibrium exhibiting one-sided strategic substitutability. We therefore position the paper as a narrow transport/spatial-competition contribution, not as a general theorem about network externalities, vertical markets, or capacity allocation.

8 Conclusion

This paper asks how spatial price competition changes when destination demand reallocates a scarce transport-service resource. The key feedback is two-sided: a retailer's price affects shopping demand directly, and that demand shifts service toward its own direction and away from its rival. Directional background demand makes the feedback asymmetric.

The model shows that this mechanism can support a nonempty open set of global price equilibria in which one retailer treats the rival's price as a strategic substitute while the other retains the conventional complementary response. A minimum service obligation can remain slack at that equilibrium yet matter for global support by truncating extreme off-path service withdrawal. The transport allocation problem also provides a coherent real-cost benchmark, under which decentralized pricing generally differs from the constrained-efficient allocation subject to the same service floor.

The interpretation is deliberately narrow. Strategic asymmetry, network effects, demand-frequency feedback, and minimum transit-service standards are established ideas; the contribution is their full-game interaction through a third-party shared service resource. The model is most natural for large destination zones, managed directional or shuttle services, and demand-responsive or shared-mobility systems. It does not derive an optimal service standard, subsidy, fleet size, or retailer location.

A Proofs and verification details

A.1 Operator allocation

Consider the slack version of (6). With multiplier λ on $f_L + f_R = F$, the first-order conditions are

$$-\frac{wD_L}{f_L^2} + \lambda = 0, \quad -\frac{wD_R}{f_R^2} + \lambda = 0. \quad (52)$$

Hence $f_L/f_R = \sqrt{D_L/D_R}$. Imposing the resource constraint gives

$$f_L = F \frac{\sqrt{D_L}}{\sqrt{D_L} + \sqrt{D_R}}, \quad f_R = F \frac{\sqrt{D_R}}{\sqrt{D_L} + \sqrt{D_R}}. \quad (53)$$

The objective is strictly convex over positive service levels whenever both directional demands are positive; at a zero-demand boundary, the positive-demand direction is served as much as the service floor permits, yielding the same clipped rule. With lower bounds $f_i \geq \underline{f}$, the KKT solution is (10).

A.2 Price first-order conditions and reaction slopes

On a stable slack branch, $G_q = 0$ and $g = G_{q,x} > 0$. Implicit differentiation gives (19). Therefore

$$\frac{\partial \pi_L}{\partial p_L} = x - \frac{p_L - c}{g}, \quad (54)$$

$$\frac{\partial \pi_R}{\partial p_R} = (1 - x) - \frac{p_R - c}{g}, \quad (55)$$

which yields (20). Differentiating these first-order conditions with respect to the rival price and using $dx/dp_R = 1/g$ or $dx/dp_L = -1/g$ produces (23)–(24). The denominators are the own-price second-order-condition terms in (22).

A.3 Proof of Proposition 1

Under fixed service, the waiting-cost difference is locally independent of x , so $g = 2t$ and $g' = 0$. Equations (23)–(24) then give $1/2$ for both slopes. With $M = 0$, the responsive-service equilibrium is symmetric at $x = 1/2$. Equation (18) gives $H_{xx}(1/2, 0) = 0$, again implying equal slopes of $1/2$. At that point $g = 2t - 4w/F$, so (20) gives the symmetric markup $t - 2w/F$. Finally, if $M > 0$ but service is unresponsive to retail shopping demand, the retail-demand slope again has no endogenous service curvature, implying $g' = 0$ and equal slopes. $\square$

A.4 Proof of Proposition 2

At the witness (27), the R -side service floor binds only for $x \geq 2/3$. In that region $G_{q,x} = 2$. On $[0, 2/3]$, direct differentiation gives (30); the exact positivity follows from

$$47133952^2 - 7599341955^2 > 0. \quad (56)$$

Thus the shopper/operator continuation is globally single-valued.

At $x^* = 23/40$, the exact local objects are

$$g^* = \frac{188234}{184117} > 0, \quad (57)$$

$$g'^* = -\frac{1058400000}{466368361}, \quad (58)$$

$$S_L = \frac{345013444}{466368361} > 0, \quad (59)$$

$$S_R = \frac{82553732}{27433433} > 0. \quad (60)$$

Substitution into (23)–(24) gives the reaction slopes in Table 1. The candidate prices follow from (20).

For global deviations, strict monotonicity of the continuation allows each firm's price problem to be parameterized by x . On the slack-frequency region, it is convenient to use the service share s . For $M = 2/3$,

$$x(s) = \frac{s^2 - (2/3)(1-s)^2}{s^2 + (1-s)^2}, \quad h(s) = \frac{1}{s} - \frac{1}{1-s}. \quad (61)$$

The physical slack interval corresponds to $s \in [s_0, 2/3]$, where $s_0 = s^u(0, 2/3) \approx 0.449490$. After substitution, each profit derivative is rational over the algebraic field $\mathbb{Q}(\sqrt{7599})$. An exact Sturm/root-count calculation over that field gives exactly two roots of $d\pi_L/ds$ and exactly one root of $d\pi_R/ds$ on the physical slack interval. The common root is exactly

$$s^* = \frac{149 - \sqrt{7599}}{98} = \frac{\sqrt{149}}{\sqrt{149} + \sqrt{51}}, \quad (62)$$

which maps to $x^* = 23/40$. The second L root lies in the rational interval $0.6609 < s < 0.6611$, corresponding to $x \approx 0.652879$. Exact derivative-sign evaluations before, between, and after these isolated roots show that x^* is a local maximum for both retailers, while the second L stationary point is a local minimum. Hence there is no omitted slack-region local maximum.

For L , the only remaining competitor is the floor boundary, and the exact gap is (31). In the binding-floor region $x \geq 2/3$, L 's profit is a concave quadratic whose vertex lies below $2/3$, so the region maximum is the boundary already compared. For R , the binding-region profit is a concave

quadratic with an interior vertex; its value is below the candidate by a strict positive amount. The corner profits are also below the candidate. Hence both firms are playing global best responses.

To establish neighborhood persistence, define the candidate equilibrium equation

$$K(x; \theta) = (2x - 1)g(x; \theta) + t(2x - 1) + AH(x, M) = 0.$$

At the witness,

$$K_x = 3g + (2x - 1)g' = \frac{1271630166}{466368361} > 0.$$

The implicit function theorem therefore gives a locally smooth candidate equilibrium. Strict floor slackness, uniform positivity of the continuation slope on each piece, own-price second-order conditions, and opposite reaction signs persist. The strict second-order conditions exclude improving deviations in a neighborhood of the candidate. On the compact complement in share space, uniqueness of each global maximizer at the witness gives a strictly positive profit gap. Continuity of the clipped allocation, including its moving boundary, preserves that gap. Corner price deviations are bounded by their endpoint profits. Thus global optimality persists on an open neighborhood; no persistence of a particular stationary-point count is required. $\square$

The exact algebra is in `verification/symbolic/stage5u_minimum_service_floor.py`. The referee-stage independent root-count, stationary-point classification, binding-region, and corner re-audit is reproduced by `verification/symbolic/stage11u_global_deviation_reaudit.py`. The deterministic neighborhood audit is reproduced by `verification/numerical/stage5u_neighborhood_audit.py`.

A.5 Proof of Proposition 3

At the witness, the equilibrium service share satisfies $q < s^* < 1 - q$, so the floor does not enter the on-path demand equation or the retail first-order conditions. Varying q locally therefore leaves the candidate allocation unchanged until an off-equilibrium continuation reaches a clipping boundary. The unilateral deviation problems are piecewise algebraic in q . Equating the candidate profit to the relevant L boundary deviation and to the relevant R binding-region maximum gives the two witness-specific roots reported in (33). Numerical root solving gives the displayed decimal endpoints; strict inequalities hold between them, and the candidate remains on the slack branch throughout that interval. The decimal endpoints are not used as universal analytic thresholds. $\square$

A.6 Proofs of Propositions 4 and 5

Substituting the unconstrained operator allocation into aggregate waiting cost gives (40). Differentiating directly yields

$$J'(x) = A \left[\frac{1}{s^u(x, M)} - \frac{1}{1 - s^u(x, M)} \right] = AH(x, M), \quad (63)$$

which proves Proposition 4. Adding spatial cost yields (42). At a retail price equilibrium, (20) implies

$$p_L - p_R = (2x - 1)g = (2x - 1)C''(x). \quad (64)$$

Combining this with (43) gives (44). If $C''' > 0$, the sign statements in Proposition 5 follow immediately. $\square$

For the witness-specific constrained-efficient allocation, when the right service floor binds, differentiating (39) gives a linear first-order condition whose solution is (46). Direct comparison

with the slack and boundary regions establishes the reported global same-floor minimizer. These calculations and the support-band roots are reproduced by `verification/symbolic/stage7u_welfare_generality.py`.

A.7 Power waiting costs

For $a(f) = wf^{-\rho}$, the operator first-order conditions imply

$$\frac{f_L}{f_R} = \left(\frac{D_L}{D_R} \right)^{1/(\rho+1)}, \quad (65)$$

which gives (49). Substitution into the objective yields J_ρ . The envelope theorem then gives (51). Because the decisive Stage-5 inequalities are strict at $\rho = 1$, continuity establishes local structural robustness. The numerical interval reported in Section 6 is a deterministic audit result rather than an analytic theorem and is reproduced by `verification/numerical/stage7u_power_waiting_robustness.py`.

Computational verification provenance

The symbolic and numerical verification scripts were developed with assistance from ChatGPT (OpenAI) and were subsequently reviewed by the author. Exact claims used in the propositions are enforced by symbolic identities, algebraic root counts, sign checks, and assertion-based tests in the reproducibility suite; numerical audits are labeled as supporting evidence rather than analytic proofs. The full verification gate is executed by `make verify`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI) to assist with literature organization, algebraic checking, drafting of verification code, and manuscript editing. The author independently reviewed and verified the mathematical derivations, references, code outputs, and manuscript text and takes full responsibility for the content of the article.

Data availability

No empirical data were used for the research described in this article. Symbolic and numerical verification materials are documented in the reproducibility package accompanying the manuscript.

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